

Supplementary material

S1 - Computation of derivatives

Let's remind $\mathcal{L}_1$, the loss function with the explicit dependence $\lambda^*(\mu)$

$$\mathcal{L}_1(\mu) = \mathcal{L}(\lambda^*(\mu), \mu) = -\alpha\Gamma(\lambda^*(\mu); \mu) + \beta R_1(\phi) + \gamma R_2(\theta) \quad (1)$$

where we introduced the shortcut $\mu = (\phi, \theta)$. The optimal parameters are

$$\mu^* = \arg \min_{\mu} \left(-\alpha \min_{\lambda} \Gamma(\lambda; \mu) + \beta R_1(\phi) + \gamma R_2(\theta) \right); \quad (2)$$

$$\lambda^* = \arg \min_{\lambda} \Gamma(\lambda; \mu^*). \quad (3)$$

In order to search for the optimal coefficients λ^*, μ^* , we rely on the BFGS algorithm (or L-BFGS-B if there are boundaries on λ , due to min, max in the χ^2 , like for *unobserved NOEs* data) implemented by Scipy. This requires the computation of the derivatives, which we can carry out by leveraging the automatic differentiation performed by JAX. However, $\mathcal{L}_1$ depends on μ also through λ^* , namely through the full minimization algorithm employed for Γ . To overcome this issue, because λ^* is a point of minimum for $\mathcal{L}$, the derivative simplifies to

$$\frac{\partial \mathcal{L}_1(\mu)}{\partial \mu_j} = \frac{\partial \mathcal{L}(\lambda, \mu)}{\partial \lambda_k} \Big|_{\lambda^*(\mu)} \frac{\partial \lambda_k^*(\mu)}{\partial \mu_j} + \frac{\partial \mathcal{L}(\lambda, \mu)}{\partial \mu_j} \Big|_{\lambda^*(\mu)} = \frac{\partial \mathcal{L}(\lambda, \mu)}{\partial \mu_j} \Big|_{\lambda^*(\mu)} \quad (4)$$

and so it can be implemented via automatic differentiation of `loss_function` with given λ^* passed as `fixed_lambdas`.

Our aim is to compute how the χ^2 in validation, evaluated at the optimal parameters λ^* and μ^* found in training, varies with the hyperparameters α, β, γ , and to compute its derivatives. By employing the chain rule, this task can be split into two parts: compute separately the derivatives of the χ^2 w.r.t. the optimal coefficients λ^*, μ^* and those of the optimal coefficients λ^*, μ^* w.r.t. the hyperparameters, then multiply opportunely the derivatives. For the sake of clarity, in the following we will use the Einstein notation (sum over repeated index is implicit).

Let's focus firstly on the second part and let's make the dependence on the hyperparameters explicit: our aim is to compute the derivatives of the optimal parameters λ^* and μ^* w.r.t. the

hyperparameters α, β, γ . The optimal coefficients are determined by the equations

$$\lambda^*(\mu; \alpha) = \arg \min_{\lambda} \Gamma(\lambda; \mu, \alpha); \quad (5)$$

$$\mu^*(\alpha, \beta, \gamma) = \arg \min_{\mu} \mathcal{L}_1(\mu; \alpha, \beta, \gamma). \quad (6)$$

In absence of boundaries, the optimal parameters are interior to the domain of the loss function. In such points the gradient is zero:

$$\left. \frac{\partial \Gamma(\lambda; \mu, \alpha)}{\partial \lambda} \right|_{\lambda^*} = 0 \text{ and } \left. \frac{\partial \mathcal{L}_1(\mu; \alpha, \beta, \gamma)}{\partial \mu} \right|_{\mu^*} = 0. \quad (7)$$

Hence, we can exploit the implicit function theorem and get to the following relations:

$$\frac{\partial \lambda_i^*}{\partial (\mu_j, \alpha)} = -(H_{in}^{-1})_{ik} \frac{\partial^2 \Gamma(\lambda; \mu, \alpha)}{\partial \lambda_k \partial (\mu_j, \alpha)} \quad (8)$$

$$\frac{\partial \mu_i^*}{\partial (\alpha, \beta, \gamma)} = -(H^{-1})_{ij} \frac{\partial^2 \mathcal{L}_1(\mu; \alpha, \beta, \gamma)}{\partial \mu_j \partial (\alpha, \beta, \gamma)} \quad (9)$$

where H_{in} is the Hessian of the internal minimization (over λ)

$$H_{inab} = \frac{\partial^2 \Gamma(\lambda; \mu, \alpha)}{\partial \lambda_a \partial \lambda_b} \quad (10)$$

and H is the Hessian of the minimization over μ

$$H_{ab} = \frac{\partial^2 \mathcal{L}_1(\mu; \alpha, \beta, \gamma)}{\partial \mu_a \partial \mu_b}. \quad (11)$$

The internal Hessian H_{in} can be directly computed through automatic differentiation of $\Gamma(\lambda; \mu, \alpha)$ by JAX (which can be equivalently done for `gamma_function` or for `loss_function`/ α with `fixed_lambdas` not None), while for the second Hessian H a further calculation is required. Indeed, we have to consider also the dependence of $\mathcal{L}_1$ on μ through λ^* . Consequently, the Hessian results

$$\begin{aligned} \frac{\partial^2 \mathcal{L}_1}{\partial \mu_j \partial \mu_k} &= \left[\frac{\partial}{\partial \lambda_l} \left(\frac{\partial \mathcal{L}}{\partial \mu_j} \right) \right]_{\lambda^*} \frac{\partial \lambda_l^*}{\partial \mu_k} + \frac{\partial^2 \mathcal{L}}{\partial \mu_j \partial \mu_k} \\ &= +\alpha \frac{\partial^2 \Gamma}{\partial \mu_j \partial \lambda_l} (H_{in}^{-1})_{lm} \frac{\partial^2 \Gamma}{\partial \lambda_m \partial \mu_k} + \frac{\partial^2 \mathcal{L}}{\partial \mu_j \partial \mu_k} \end{aligned} \quad (12)$$

where the first term has to be summed over the molecular systems used in cross validation, which are possibly different from those used in training. Similarly, it holds

$$\frac{\partial^2 \mathcal{L}_1}{\partial \mu_j \partial \alpha} = \frac{\partial}{\partial \alpha} \frac{\partial \mathcal{L}(\lambda^*(\mu, \alpha), \mu; \alpha, \beta, \gamma)}{\partial \mu_j} = \frac{\partial^2 \mathcal{L}}{\partial \mu_j \partial \lambda_k} \frac{\partial \lambda_k}{\partial \alpha} + \frac{\partial^2 \mathcal{L}}{\partial \mu_j \partial \alpha}. \quad (13)$$

Also these terms can be computed through automatic differentiation.

Then, let's focus on the first part, namely, the derivatives of the χ^2 with respect to the optimal parameters λ^*, μ^* . The full dependence is through the average value of the validating observables $\tilde{g}_i(\theta^*)$ (computed with the optimized forward model θ^*) on the ensemble corresponding to the optimal parameters λ^*, μ^* :

$$\langle \tilde{g}_i(\theta^*(\alpha, \beta, \gamma)) \rangle_{\lambda^*(\mu^*(\alpha, \beta, \gamma); \alpha), \mu^*(\alpha, \beta, \gamma)}. \quad (14)$$

These derivatives can be directly computed through automatic differentiation. Let's conclude our calculation by applying the chain rule:

$$\frac{\partial \chi^2}{\partial \alpha} = \frac{\partial \chi^2}{\partial \mu_i} \frac{\partial \mu_i^*}{\partial \alpha} + \frac{\partial \chi^2}{\partial \lambda_i} \left(\frac{\partial \lambda_i^*}{\partial \mu_j} \frac{\partial \mu_j^*}{\partial \alpha} + \frac{\partial \lambda_i^*}{\partial \alpha} \right) \quad (15)$$

$$\frac{\partial \chi^2}{\partial (\beta, \gamma)} = \frac{\partial \chi^2}{\partial \mu_i} \frac{\partial \mu_i^*}{\partial (\beta, \gamma)} + \frac{\partial \chi^2}{\partial \lambda_i} \frac{\partial \lambda_i^*}{\partial \mu_j} \frac{\partial \mu_j^*}{\partial (\beta, \gamma)}. \quad (16)$$

Now, each term on the right side of the equations can be computed through automatic differentiation and we just have to multiply them as indicated in order to get the derivatives of χ^2 w.r.t. the hyperparameters α, β, γ . This will facilitate the search for the optimal hyperparameters, without the need for a scan in 3 dimensions, which may be cumbersome. Lastly, the minimization can be sped up considering the fact that one needs to explore several orders of magnitude in the hyperparameter space. It is then convenient to change variable and perform the minimization considering as hyperparameters $\log \alpha$, $\log \beta$, and $\log \gamma$, using once again the chain rule for derivatives $\frac{\partial}{\partial \log \alpha} = \alpha \frac{\partial}{\partial \alpha}$, and analogous expressions for the other hyperparameters.

These calculations can be adapted to the presence of boundaries for the λ parameters (like in the case of *unobserved NOEs*). In these cases, when the optimal solution is on the boundary $\lambda_j = 0$ for some j , the derivatives of these λ_j parameters w.r.t. α, μ is zero. To prove this statement, let's consider for simplicity the case with the constraint $\lambda_j > 0$ for some j . When the optimal solution is on the boundary $\lambda_j = 0$ it means that, if we could remove the boundary, the minimum of the loss function $\Gamma(\lambda; \alpha, \mu)$ would be inner to the forbidden region $\lambda_j < 0$. Hence, by continuity of the Γ function with respect to α (such as for μ), when varying α (or μ) by an arbitrarily small value, the minimum will still be in the forbidden region and the value of λ_j will still be zero, without changing. This holds for every function $f(\lambda; \alpha, \mu)$ which is continuous in its arguments, independently of its convexity or not. Notice that since the boundaries are in the form of linear inequalities on λ , the function Γ is expected to be convex also on the boundary.

S2 - Implementation of minimizer

To determine the optimal $\lambda^*(\phi, \theta)$ at given (ϕ, θ) we can employ the same `gamma_function` previously described. Notice `gamma_function` does not include either the computation of the observables through the forward model θ or the reweighting through the force-field correction ϕ (namely, the dependence of Γ on θ, ϕ), rather they should be externally computed and then passed to `gamma_function` as input observables and weights.

Regarding the full loss function, we implemented $\mathcal{L}_1(\phi, \theta)$ in `loss_function` by inserting the search for optimal $\lambda^*(\phi, \theta)$. Its inputs are the Numpy array of (ϕ, θ) coefficients, the data object, the regularization dictionary, the hyperparameters α, β, γ (by default $+\infty$, i.e., no refinement in that direction) and some other variables. The regularization dictionary has the two keys `force_field_reg`, `forward_model_reg`; the first item can be either a string (KL divergence, plain l2) or a user-defined function acting on ϕ ; the second item is a user-defined function acting on θ and θ_0 (for example, an L^2 regularization).

The other inputs include: `fixed_lambdas` (by default None), `gtol_inn` and `bounds` (tolerance and boundaries for the inner minimization of λ ; default values 10^{-3} and None, respectively), `if_save` (Boolean variable, by default False). For `if_save = True`, `loss_function` does not return just its numerical value, rather an object with all the quantities appearing in Eq. 9 of the Main text, such as the parameters (λ, ϕ, θ) (when defined), the new weights, the χ^2 , the regularization terms.

Whenever the input variable `fixed_lambdas` is not None, `loss_function` implements the computation of $\mathcal{L}_1(\phi, \theta)$ through substitution of the given λ into $\Gamma(\lambda; \phi, \theta)$ in place of the inner minimization of Γ over λ (the two computations are the same if α is infinite, namely, no ensemble refinement). This distinction turns out to be crucial when computing the gradient function $\frac{\partial \mathcal{L}_1(\phi, \theta)}{\partial(\phi, \theta)}$ `gradient_fun` with automatic differentiation (see *Supplementary - Computation of derivatives*).

The `loss_function` and its `gradient_fun` are evaluated together inside `loss_function_and_grad`, which is employed in `minimizer` in order to get the optimal solution. In this way, passing to `minimizer` the data object, the regularization dictionary and the hyperparameters (by default $\alpha, \beta, \gamma = +\infty$, i.e., no refinement in those directions), we get an output object `Result` containing the optimal solution (P, ϕ, θ) and its properties.

S3 - Kish sample size and entropy

Let's consider a sample $\{x_1, \dots, x_n\}$ (assumed to be uncorrelated) with corresponding normalized weights $\{w_1, \dots, w_n\}$. If the weights are uniform, then the effective sample size is simply given by the number of points $n_{eff} = n$. Otherwise, we can obtain a good estimation of the effective sample size by using the Kish sample size, defined as the inverse of the average weight

$$n_{eff} = K[\{w_i\}] := \langle w \rangle_w^{-1} = \left(\sum_i w_i^2 \right)^{-1}, \quad (17)$$

This definition of n_{eff} reduces to $n_{eff} = n$ if each sample contributes with uniform weight, while it returns the number of "dominant" frames if some of them count much more than the others. For example, if we have a non-normalized set of weights $\{1, \dots, 1, N, \dots, N\}$ with weights 1 and N repeated $n - m$ and m times, respectively, then the effective sample size is $n_{eff} \simeq m$ if $n - m \ll Nm$ (the subset of small weights $w_i = 1$ dominates over the other) and $n_{eff} \simeq n - m$ in the opposite limit. Clearly, it is always $n_{eff} \leq n$ and the equality holds only for uniform weights; it is also $n_{eff} > 1$ and small values are achieved when one frame contributes much more than the others.

The relation with the entropy S is straightforward. It suffices to change the weighted arithmetic average $\langle \dots \rangle$ to the weighted geometric one in Eq. 17 to get

$$e^{S[\{w_i\}]} = \left(\prod_i w_i^{w_i} \right)^{-1}, \quad (18)$$

which is always in between $1 < e^{S[\{w_i\}]} \leq n$. Hence, the effective number of frames can be estimated also through $e^{S[\{w_i\}]}$, in a way which could be less sensitive to the fact that weights could vary over orders of magnitude. Moreover, by the Jensen's inequality, it is always $e^{S[\{w_i\}]} \geq K[\{w_i\}]$.

Then, the relative entropy S_{rel} can be used to quantify the discrepancy of the distribution P , represented by the set of weights $\{w_i\}$, from P_0 , represented by $\{w_{0i}\}$:

$$e^{S_{rel}[P|P_0]} = \left[\prod_i \left(\frac{w_i}{w_{0i}} \right)^{w_i} \right]^{-1}. \quad (19)$$

The relative entropy is always $S_{rel} \leq 0$ but can be very large in absolute value for P very different from P_0 , with a lower bound determined by the number of frames. Notice it is the geometric average (over w) of the ratio w/w_0 ; changing it to the arithmetic average we get the relative Kish size¹

$$K_{rel}[P|P_0] = \left\langle \frac{w}{w_0} \right\rangle_w^{-1}. \quad (20)$$

If the reference ensemble is uniform ($w_0 = \frac{1}{n}$ constant), then the relative Kish size (and analogously $e^{S_{rel}[P|P_0]}$) reduces to the ratio n_{eff}/n .

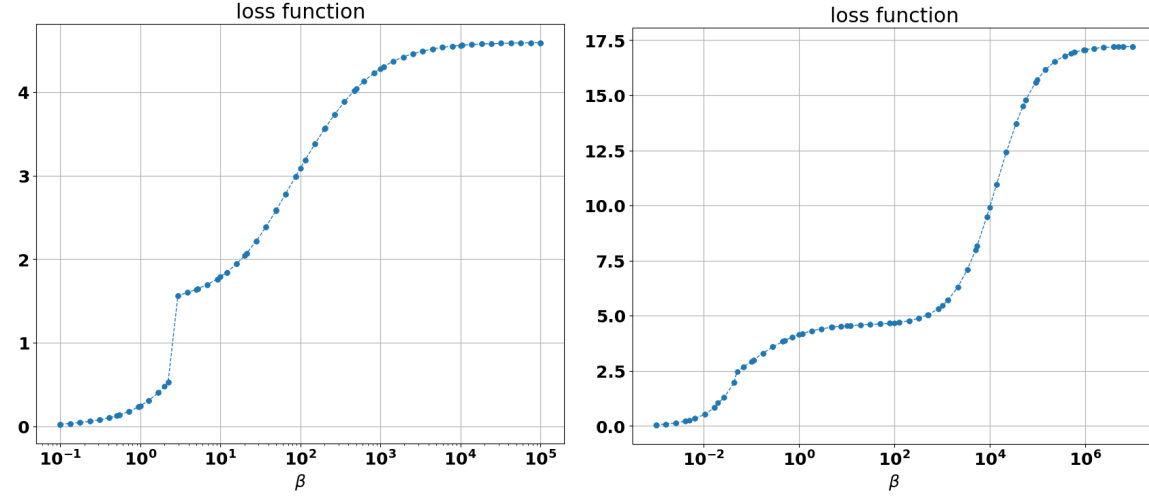
Finally, the Kish size ratio (KSR) for an ensemble P reweighted from P_0 has been defined² as the ratio between the two Kish sample sizes of P and P_0 , respectively:

$$KSR = \frac{\langle w \rangle_w^{-1}}{\langle w_0 \rangle_{w_0}^{-1}}. \quad (21)$$

By following the analogy between Kish size and entropy, which share the same definition but using arithmetic and geometric averages, respectively, the Kish size ratio can be considered analogous to the exponential of the entropy difference $e^{S[P]-S[P_0]}$.

S4 - Force-field refinement with alchemical calculations

(a) min. loss function for L^2 regularization; the overfitting transition is quite evident (b) min. loss function for relative entropy regularization



(c) χ^2 for different thermodynamic cycles and average KSR for L^2 regularization (d) χ^2 for different thermodynamic cycles and average KSR for relative entropy regularization

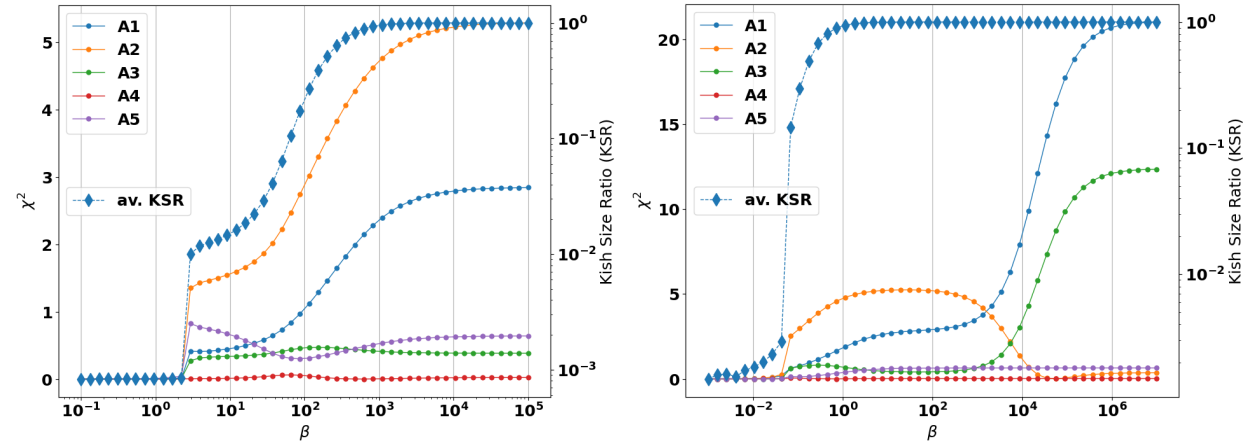


FIG. S1: Loss function, average Kish size ratio and χ^2 for each thermodynamic cycle at the optimal solution found at given hyperparameter β and regularization (L^2 on charges or relative entropy). The overfitting transition is quite evident in the case of L^2 regularization, while it is smoother for the relative entropy regularization.

In Fig. S1 we report the minimum value of the loss function, the χ^2 for the different thermodynamic cycles and the average Kish size ratio (KSR) as a function of the hyperparameter β . The overfitting transition is quite evident in the case with L^2 regularization, while it is smoother in the second case. For the sake of clarity, in section S2, we report the definitions of Kish size ratio and relative Kish size, introduced in previous works^{2,3}, and highlight their relationships with the entropy or the relative entropy.

S5 - Forward-model refinement + max. entropy

In Table S1 we report the optimized empirical coefficients of the Karplus equations, comparing them with values already known in the literature. In Fig. S2 we report the Refinement of the empirical coefficients for the Karplus equations with Ensemble + Forward Model Refinement.

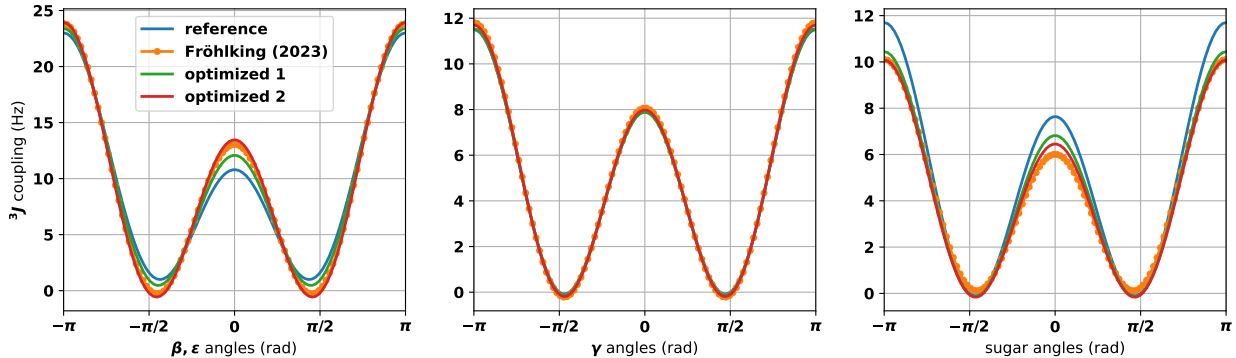


FIG. S2: Refinement of the empirical coefficients for the Karplus equations with Ensemble + Forward Model Refinement; comparison between *optimized 1* and *optimized 2* (see Table S1).

In Fig. S3 we report the Kullback-Leibler divergence of the refined ensembles with respect to the original ones, represented by MD simulations. These values are quite small: the ensembles do not change significantly; this is confirmed by Fig. S4 showing weighted histograms of some dihedral angles.

name	A	B	C
β, ε			
reference	15.3	-6.1	1.6
Fröhlking	18.34	-5.39	0.11
optimized 1	16.80	-5.66	0.94
optimized 2	18.91	-5.23	-0.21
fully combined	18.13	-5.29	0.072
γ			
reference	9.7	-1.8	0
Fröhlking	10.07	-1.87	-0.13
optimized 1	9.77	-1.83	-0.057
optimized 2	9.95	-1.86	-0.11
fully combined	9.92	-1.76	-0.12
<i>sugar</i>			
reference	9.67	-2.03	0
Fröhlking	7.81	-2.05	0.25
optimized 1	8.68	-1.81	-0.049
optimized 2	8.34	-1.81	-0.069
fully combined	6.99	-1.64	0.67

TABLE S1: Empirical coefficients A, B, C (in Hz) of the Karplus equations. Comparison between: reference values; coefficients determined by Fröhlking et al. in³ (hyperparameters $\alpha = 174.33288$, $\gamma = 6.86664/2Hz^{-2}$ - factor 1/2 because of different regularization prefactor); coefficients determined in this work (optimized 1, i.e., with same hyperparameters as in³; optimized 2, i.e., with optimal hyperparameters found in this work $\alpha = 46.31$, $\gamma = 0.96Hz^{-2}$); coefficients determined in this work with the fully combined approach ($\alpha = 959.86$, $\beta = 167.94$, $\gamma = 0.42Hz^{-2}$).

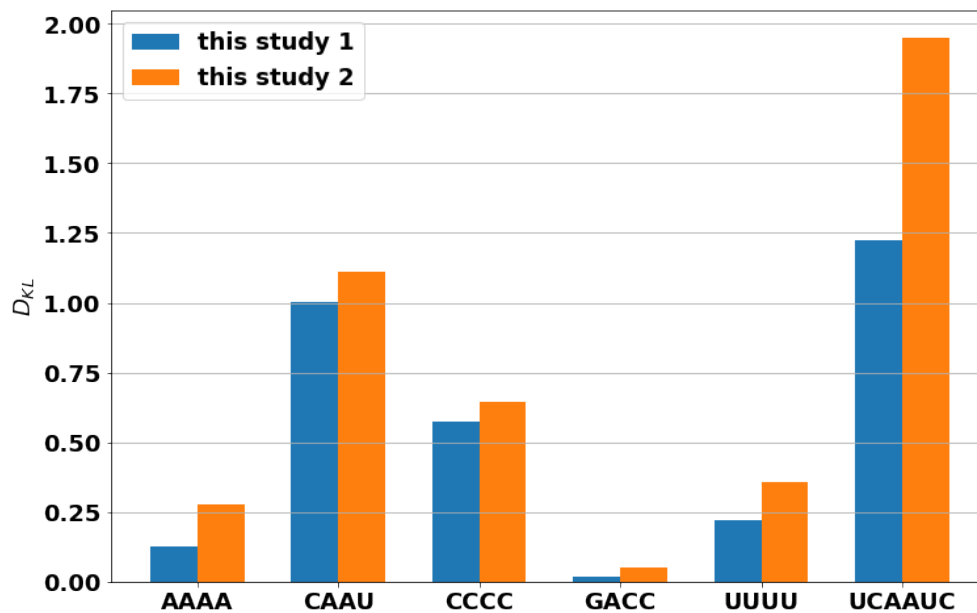


FIG. S3: Refinement of Karplus equations: Kullback-Leibler divergences of the refined ensembles with respect to the original ones (optimized 1 with same hyperparameters as in³; 2 with optimized hyperparameters).

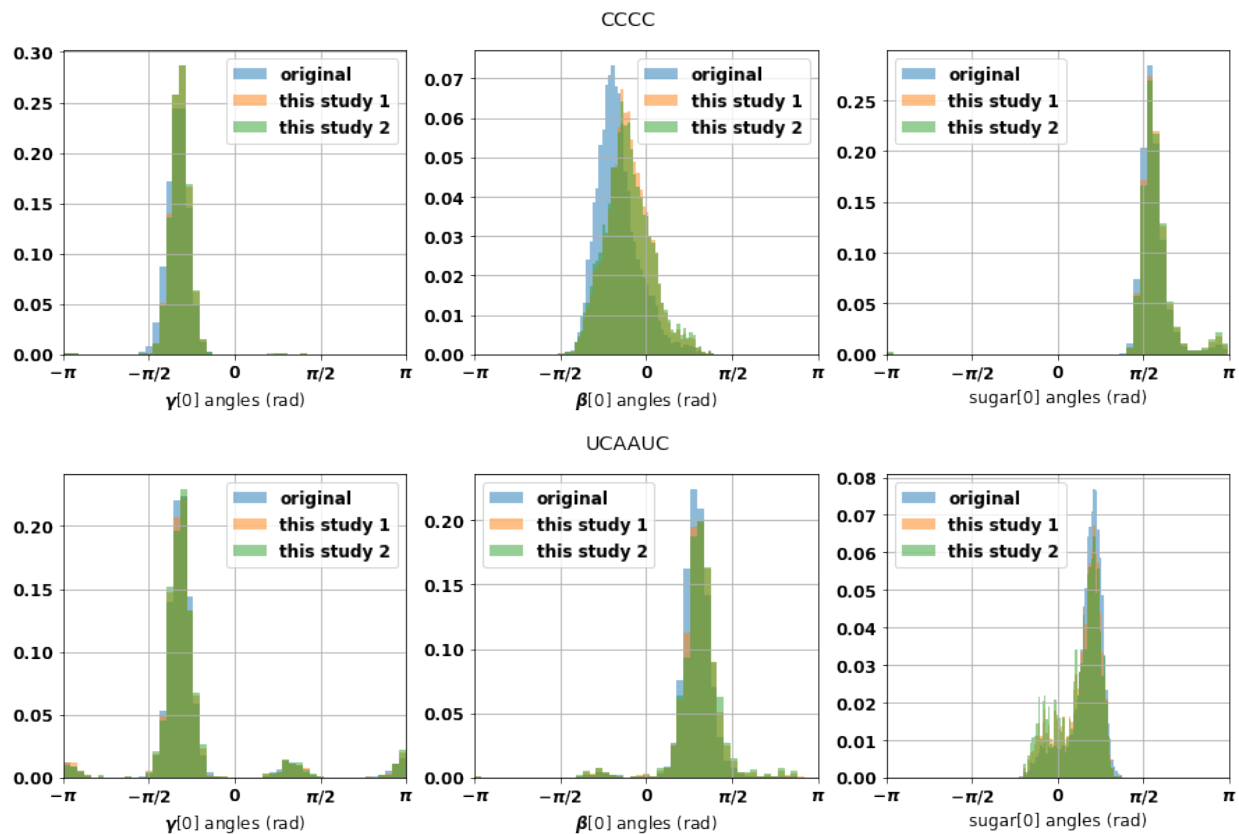


FIG. S4: Weighted histograms of some dihedral angles of CCCC and UCAAUC oligomers, as examples (comparison between original ensemble and reweighted ones).

S6 - Fully combined refinement

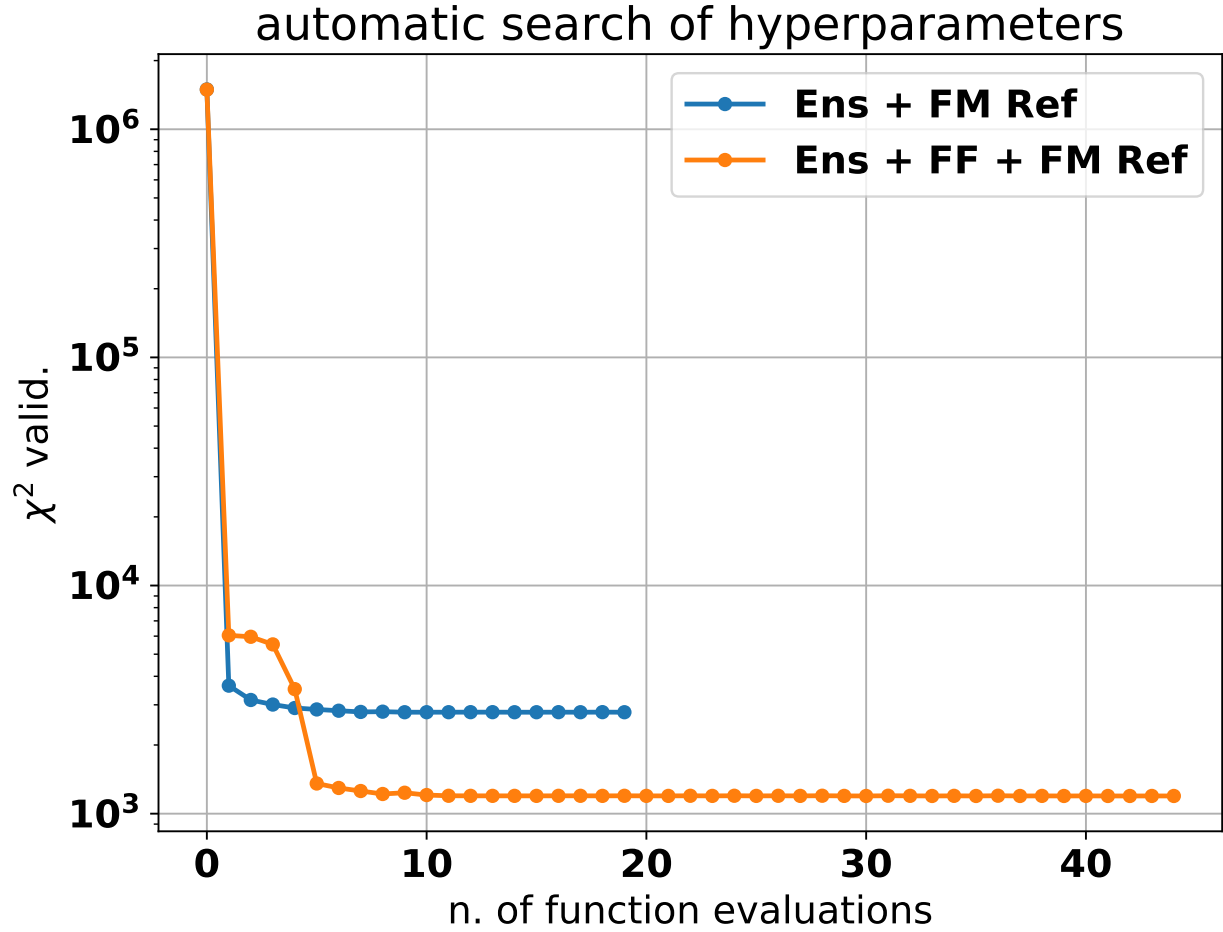


FIG. S5: Average value of χ^2 on validating observables in the course of the automatic search for the optimal hyperparameters. Comparison among Ens + FM (Ensemble + Forward Model) Refinement and Ens + FF + FM (the fully combined) Refinement. Starting from the initial values $\alpha = \beta = 1, \gamma = 1 Hz^{-2}$, we get the final values $\chi^2 = 2784.1$ for Ens+FM Ref. ($\alpha = 46.31, \gamma = 0.96 Hz^{-2}$) and $\chi^2 = 1195.6$ for the fully combined approach ($\alpha = 959.9, \beta = 167.9, \gamma = 0.42 Hz^{-2}$). The cross validation is performed on the full dataset (without test observables).

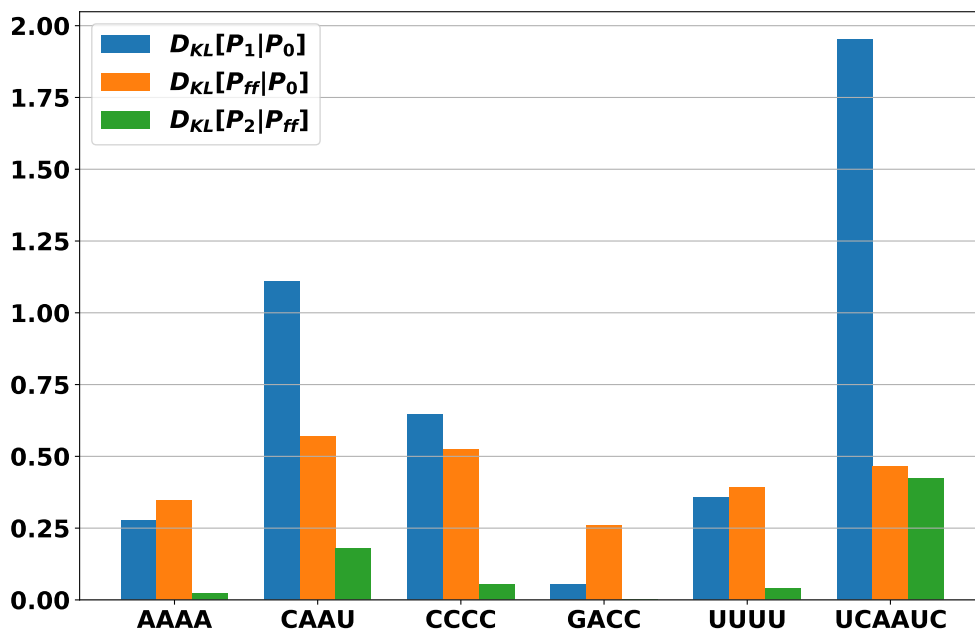


FIG. S6: Comparison of the Kullback-Leibler divergences for different molecular systems when performing E+FM refinement (P_1) or the fully combined approach, with P_{ff} the force-field correction ensemble and P_2 the final ensemble in the loss function (Eq. 9 of the Main text).

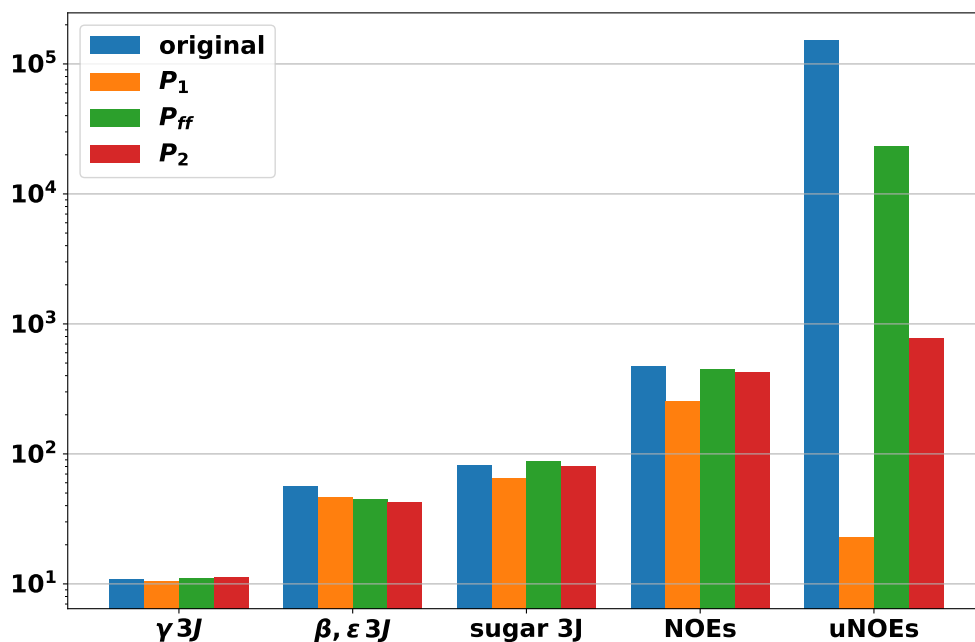


FIG. S7: Comparison of the χ^2 for different experimental measures when performing E+FM refinement (P_1) or the fully combined approach, with P_{ff} the force-field correction ensemble and P_2 the final ensemble in the loss function (Eq. 9 of the Main text) (sum over different molecular systems).

A. S7 - Nested cross-validation

We set aside a test set (not used in cross-validation) in order to report the agreement of the refined models with experiments, to assess for model evaluation. Specifically, we have used `split_dataset` to select a test set and then repeated the minimizations with and without force-field fitting on the reduced dataset in the same way described in the “Results” section of main text (namely, including the search for the optimal hyperparameters as performed by MDRefinement). The results we have determined on the subset are shown in Fig S8 and Tables S3, S2. Once the optimal solution has been found on this reduced dataset, we evaluate the quality of the two refinement methods by reporting the chi2 values (normalized to the total n. of observables) on the test set in Table S4.

	α^*	β^*	γ^*	$\sin \alpha$	$\cos \alpha$
reference	$+\infty$	$+\infty$	$+\infty$	-	-
E + FM full	46.31	$+\infty$	0.96	-	-
E + FM + FF full	959.86	167.94	0.42	0.58	-0.79
E + FM reduced	14.77	$+\infty$	0.52	-	-
E + FM + FF reduced	37.24	1.89	0.35	0.61	-1.14

TABLE S2: Optimal values of the hyperparameters $\alpha^*, \beta^*, \gamma^*$ and coefficients of the sinusoidal force-field on α dihedral angles (in $k_B T$). Comparison between: reference values; coefficients determined with/without including force-field fitting (E + FM or E + FM + FF) on the full dataset or on a reduced dataset (got by removing a test set).

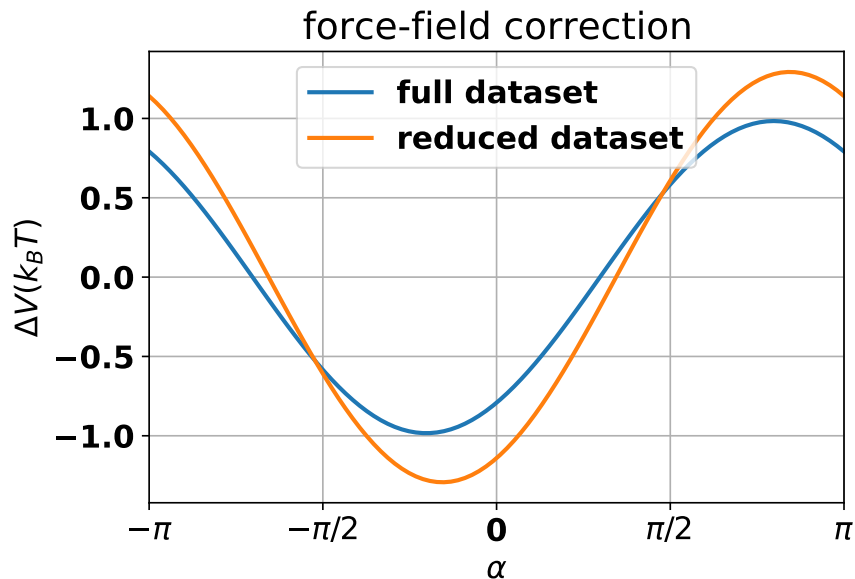
	A_β	B_β	C_β	A_γ	B_γ	C_γ	A_v	B_v	C_v
reference	15.3	-6.1	1.6	9.7	-1.8	0	9.67	-2.03	0
E + FM full	18.91	-5.23	-0.21	9.95	-1.86	-0.11	8.34	-1.81	-0.069
E + FM + FF full	18.13	-5.29	0.072	9.92	-1.76	-0.12	6.99	-1.64	0.67
E + FM reduced	19.59	-5.12	-0.54	10.47	-1.84	-0.22	8.42	-1.87	-0.12
E + FM + FF reduced	18.57	-5.22	-0.17	10.30	-1.75	-0.19	7.43	-1.72	0.42

TABLE S3: Empirical coefficients A, B, C (in Hz) of the Karplus equations. Comparison between: reference values; coefficients determined with/without including force-field fitting (E + FM or E + FM + FF) on the full dataset or on a reduced dataset (got by removing a test set).

test χ^2_{red} original	test χ^2_{red} E+FM	test χ^2_{red} E+FM+FF
537.4	1.62	2.13

TABLE S4: Reduced χ^2 (namely, χ^2 divided by the n. of observables) on the test set; original set of weights (before the refinement), refinement of ensemble and forward model (E+FM), fully combined refinement (E+FM+FF).

(a) force-field correction on α dihedral angles



(b) refinement of the Karplus equations

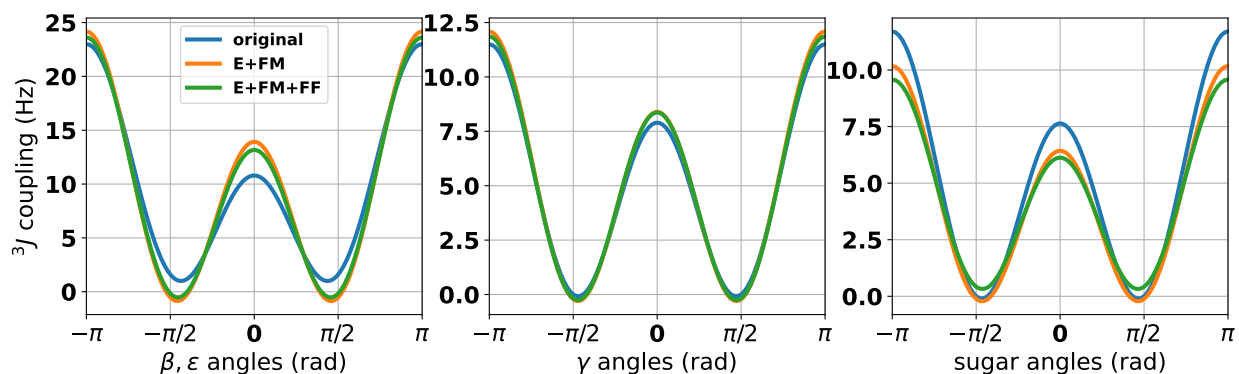


FIG. S8: Force-field correction (a) and refinement of the Karplus equations (b) derived by minimization of the loss function on a *reduced dataset*, leaving out a test set of 10% frames and 10% observables. In (a) we report also the force-field correction determined on the *full dataset*, which overlaps with the other in the visited regions, close to $\pm\pi/2$ (see also Fig. 5a in the Main text). In (b) we do not report the refinement derived with the full dataset (Main, Fig. 6) since it would completely overlap with the corresponding plotted refinements.

REFERENCES

- ¹T. Fröhlking, V. Mlýnský, M. Janeček, P. Kührová, M. Krepl, P. Banáš, J. Šponer, and G. Bussi, “Automatic learning of hydrogen-bond fixes in an AMBER RNA force field,” *J. Chem. Theory Comput.* **18**, 4490–4502 (2022).
- ²V. Pionponi, T. Fröhlking, M. Bernetti, and G. Bussi, “Molecular simulations matching denaturation experiments for N6-methyladenosine,” *ACS Cent. Sci.* **8**, 1218–1228 (2022).
- ³T. Fröhlking, M. Bernetti, and G. Bussi, “Simultaneous refinement of molecular dynamics ensembles and forward models using experimental data,” *J. Chem. Phys.* **158** (2023).